

Codebook

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This codebook describes the variables contained in the replication datasets. Sources are abbreviated as follows:

- **CPED**: Jiang, Junyan, 2018, Chinese Political Elite Database, <https://www.junyanjiang.com/data.html>
- **Face++**: Face++ API (<https://www.faceplusplus.com/>)
- **Dlib**: Computed from 68-point facial landmarks using dlib (King, 2009; Kazemi and Sullivan, 2014).
- **Survey**: Conjoint survey experiment

Table 1: all_asperson_original.dta — Pre-merge person-rank level data with facial measures and trait ratings (input to merge_cped_bio.do; merging CPED biographical variables produces all_asperson_fullldata.dta)

Variable Name	Description	Source
<i>Identifiers and Career</i>		
name	Official's anonymized ID	CPED
rank	Administrative rank	CPED
begin	Date of appointment	CPED
edu_year	Years of education	CPED
purge	Purged for corruption	CPED
latitude	Birth city latitude	
longitude	Birth city longitude	
exists_in_psub	Exists in psub dataset	
<i>Objective Facial Attributes</i>		
fwhr	Facial width-to-height ratio	Dlib
mouth_width	Mouth width	Dlib
nose_width	Nose width	Dlib
nose_length	Nose length	Dlib
nwhr	Nasal width-to-height ratio	Dlib
eyebrow_position	Eyebrow position	Dlib
foreheadr	Forehead ratio	Dlib

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Variable Name	Description	Source
lip_thickness	Lip thickness	Dlib
eyesize	Eye size	Dlib
chin_length	Chin length	Dlib
chin_width	Chin width	Dlib
flength	Face length	Dlib
<i>Image Quality and Attributes</i>		
smile_mean	Smile score	Face++
facequality_mean	Face quality	Face++
blurriness_mean	Blurriness	Face++
glass_mean	Wears glasses	Face++
facequality_median	Face quality (median)	Face++
blurriness_median	Blurriness (median)	Face++
<i>Facial Trait Ratings (Human)</i>		
attractiveness	Perceived attractiveness	Rater
competence	Perceived competence	Rater
trustworthiness	Perceived trustworthiness	Rater
aggressiveness	Perceived aggressiveness	Rater
attractiveness_median	Perceived attractiveness (median)	Rater
competence_median	Perceived competence (median)	Rater
trustworthiness_median	Perceived trustworthiness (median)	Rater
aggressiveness_median	Perceived aggressiveness (median)	Rater
<i>Facial Similarity to Leaders</i>		
similarity_xjp_all	Similarity to Xi	Face++
similarity_xjp_hs	Similarity to Xi (headshot only)	Face++
similarity_hjt_all	Similarity to Hu	Face++
similarity_hjt_hs	Similarity to Hu (headshot only)	Face++
similarity_jzm_all	Similarity to Jiang	Face++
similarity_jzm_hs	Similarity to Jiang (headshot only)	Face++

Table 2: all_asperson_fulldata.dta — Person-rank level data with facial trait ratings, demographics, and career outcomes

Variable Name	Description	Source
<i>Identifiers and Career</i>		
name	Official’s anonymized ID	CPED
rank	Administrative rank	CPED
birthyear	Birth date	CPED
begin	Date of appointment	CPED
female	Female indicator	CPED
education	Highest education	CPED
nation	Ethnicity	CPED

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Variable Name	Description	Source
birth_prov	Birth province	CPED
birth_city	Birth city	CPED
purge	Purged for corruption	CPED
latitude	Birth city latitude	
longitude	Birth city longitude	
<i>Objective Facial Attributes</i>		
fwhr	Facial width-to-height ratio	Dlib
mouth_width	Mouth width	Dlib
nose_width	Nose width	Dlib
nose_length	Nose length	Dlib
nwhr	Nasal width-to-height ratio	Dlib
eyebrow_position	Eyebrow position	Dlib
foreheadr	Forehead ratio	Dlib
lip_thickness	Lip thickness	Dlib
eyesize	Eye size	Dlib
chin_length	Chin length	Dlib
chin_width	Chin width	Dlib
flength	Face length	Dlib
<i>Image Quality and Attributes</i>		
smile_mean	Smile score	Face++
facequality_mean	Face quality	Face++
blurriness_mean	Blurriness	Face++
glass_mean	Wears glasses	Face++
facequality_median	Face quality	Face++
blurriness_median	Blurriness	Face++
<i>Facial Trait Ratings (Human)</i>		
attractiveness	Perceived attractiveness	Rater
competence	Perceived competence	Rater
trustworthiness	Perceived trustworthiness	Rater
aggressiveness	Perceived aggressiveness	Rater
attractiveness_median	Perceived attractiveness	Rater
competence_median	Perceived competence	Rater
trustworthiness_median	Perceived trustworthiness	Rater
aggressiveness_median	Perceived aggressiveness	Rater
<i>Facial Similarity to Leaders</i>		
similarity_xjp_all	Similarity to Xi	Face++
similarity_xjp_hs	Similarity to Xi	Face++
similarity_hjt_all	Similarity to Hu	Face++
similarity_hjt_hs	Similarity to Hu	Face++
similarity_jzm_all	Similarity to Jiang	Face++
similarity_jzm_hs	Similarity to Jiang	Face++
exists_in_psub	Exists in psub dataset	

Table 3: proper.dta — Auxiliary data for matching officials by name, birth year, and birth province

Variable Name	Description	Source
name	Official’s anonymized ID	all_asperson_fulldata.dta
byear	Birth year	all_asperson_fulldata.dta
birth_prov	Birth province	all_asperson_fulldata.dta
rank1	Observed at rank 1	all_asperson_fulldata.dta
rank2_in_lower	Rank 2 observed at lower level	all_asperson_fulldata.dta
rank3_in_lower	Rank 3 observed at lower level	all_asperson_fulldata.dta
rank4_in_lower	Rank 4 observed at lower level	all_asperson_fulldata.dta
rank5_in_lower	Rank 5 observed at lower level	all_asperson_fulldata.dta

Table 4: hs_mt.dta — Facial trait ratings from headshot and meeting photos separately

Variable Name	Description	Source
name	Official’s anonymized ID	all_asperson_fulldata.dta
birth_prov	Birth province	all_asperson_fulldata.dta
byear	Birth year	all_asperson_fulldata.dta
attractiveness_hs	Attractiveness	Rater
competence_hs	Competence	Rater
aggressiveness_hs	Aggressiveness	Rater
trustworthiness_hs	Trustworthiness	Rater
attractiveness_mt	Attractiveness	Rater
competence_mt	Competence	Rater
aggressiveness_mt	Aggressiveness	Rater
trustworthiness_mt	Trustworthiness	Rater

Table 5: zzj_asimage_face++.dta — Image-level data for Politburo members across career stages

Variable Name	Description	Source
<i>Identifiers</i>		
name	Official’s anonymized ID	all_asperson_fulldata.dta
rank_pic_taken	Rank when photo was taken	Original
year_pic_taken	Year photo was taken	Original
birth_year	Birth year	all_asperson_fulldata.dta
gender	Gender	all_asperson_fulldata.dta
<i>Facial Trait Ratings (Human)</i>		
attractiveness	Perceived attractiveness	Rater
competence	Perceived competence	Rater
trustworthiness	Perceived trustworthiness	Rater
aggressiveness	Perceived aggressiveness	Rater

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Variable Name	Description	Source
<i>Face++ Attributes</i>		
blurriness	Blurriness	Face++
facequality	Face quality score	Face++

Table 6: all_asperson_gdpconn_undemean.dta — Person-level data with economic and fiscal performance indicators

Variable Name	Description	Source
<i>Identifiers and Demographics</i>		
name	Official’s anonymized ID	all_asperson_fulldata.dta
rank	Administrative rank	all_asperson_fulldata.dta
birthyear	Birth date	all_asperson_fulldata.dta
begin	Date of appointment	all_asperson_fulldata.dta
female	Female indicator	all_asperson_fulldata.dta
education	Highest education	all_asperson_fulldata.dta
edu	Education level	all_asperson_fulldata.dta
edu_year	Years of education	all_asperson_fulldata.dta
nation	Ethnicity	all_asperson_fulldata.dta
eth_han	Han ethnicity indicator	all_asperson_fulldata.dta
birth_prov	Birth province	all_asperson_fulldata.dta
birth_city	Birth city	all_asperson_fulldata.dta
purge	Purged for corruption	all_asperson_fulldata.dta
latitude	Birth city latitude	all_asperson_fulldata.dta
longitude	Birth city longitude	all_asperson_fulldata.dta
connections	Political connections	Original
<i>Objective Facial Attributes</i>		
fwhr1	Facial width-to-height ratio	Dlib
mouth_width	Mouth width	Dlib
nose_width	Nose width	Dlib
nose_length	Nose length	Dlib
nwhr	Nasal width-to-height ratio	Dlib
eyebrow_position	Eyebrow position	Dlib
foreheadr	Forehead ratio	Dlib
lip_thickness	Lip thickness	Dlib
eyesize	Eye size	Dlib
chin_length	Chin length	Dlib
chin_width	Chin width	Dlib
flength	Face length	Dlib

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Variable Name	Description	Source
<i>Facial Trait Ratings</i>		
attractiveness	Perceived attractiveness	Rater
competence	Perceived competence	Rater
trustworthiness	Perceived trustworthiness	Rater
aggressiveness	Perceived aggressiveness	Rater
<i>Economic Performance</i>		
gdp	GDP (demeaned)	Original
weighted_avg_gdp_growth_rate	Weighted avg. GDP growth rate	Original
weighted_avg_gdp	Weighted avg. GDP	Original
weighted_avg_fiscal_growth_rate	Weighted avg. fiscal revenue growth	Original
weighted_avg_fiscal_revenue	Weighted avg. fiscal revenue	Original
avg_gdp_growth_rate_t_plus_2	Avg. GDP growth at $t+2$	Original
avg_fiscal_growth_rate_t_plus_2	Avg. fiscal revenue growth at $t+2$	Original

Table 7: validation_rep.dta — Conjoint survey representativeness validation data

Variable Name	Description	Source
province	Province name	
n	Number of survey respondents	
year	Year	
new_recruits	Civil service recruitment size	
log_n	Log number of respondents	
log_recruit	Log recruitment size	
_est_m1	Estimation-sample indicator from <code>estimates store m1</code> (equals 1 for observations used in the validation regression)	Analysis

Table 8: 10fold_training_results.csv — Per-observation predictions from 10-fold cross-validation of the human-rated facial trait model (used for Figure 2)

Variable Name	Description	Source
ImagePath	Image file path	
TrueScore	Human-rated score	Rater

PredictedScore trait	Model predicted score Facial trait (attractiveness, competence, trustworthiness, aggressiveness)	Analysis
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Table 9: dis_promo.csv — Marginal effects from discrete (quartile-based) facial trait measures for promotion outcome

Variable Name	Description	Source
beta	Coefficient estimate	Analysis
se	Standard error	Analysis
lb95	Lower bound (95% CI)	Analysis
ub95	Upper bound (95% CI)	Analysis
lb90	Lower bound (90% CI)	Analysis
ub90	Upper bound (90% CI)	Analysis
rn	Variable name (e.g., "1bn.attractiveness_4l", "2.attractiveness_4l", etc.)	

Table 10: dis_purge.csv — Marginal effects from discrete (quartile-based) facial trait measures for purge outcome

Variable Name	Description	Source
beta	Coefficient estimate	Analysis
se	Standard error	Analysis
lb95	Lower bound (95% CI)	Analysis
ub95	Upper bound (95% CI)	Analysis
lb90	Lower bound (90% CI)	Analysis
ub90	Upper bound (90% CI)	Analysis
rn	Variable name (e.g., "1bn.attractiveness_4l", "2.attractiveness_4l", etc.)	

Table 11: promotion_results.csv — Feature importance from random forest for the promotion outcome (used for Figure 4)

Variable Name	Description	Source
Feature	Feature name (e.g., competence, trustworthiness, aggressiveness, attractiveness, weighted_avg_gdp_growth_rate, latitude, etc.)	

Importance	Feature importance score from random forest model	Analysis
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Table 12: purge_results.csv — Feature importance from random forest for the purge outcome (used for Figure 4)

Variable Name	Description	Source
Feature	Feature name (e.g., weighted_avg_fiscal_growth_rate, weighted_avg_gdp_growth_rate, latitude, birthyear, aggressiveness, etc.)	
Importance	Feature importance score from random forest model	Analysis

Table 13: attractiveness.csv — Exported marginal effects from the conjoint regression for attractiveness

Variable Name	Description	Source
beta	Coefficient estimate	Analysis
se	Standard error	Analysis
lb95	Lower bound (95% CI)	Analysis
ub95	Upper bound (95% CI)	Analysis
lb90	Lower bound (90% CI)	Analysis
ub90	Upper bound (90% CI)	Analysis
rn	Variable name (treatment indicator)	

Table 14: competence.csv — Exported marginal effects from the conjoint regression for competence

Variable Name	Description	Source
beta	Coefficient estimate	Analysis
se	Standard error	Analysis
lb95	Lower bound (95% CI)	Analysis
ub95	Upper bound (95% CI)	Analysis
lb90	Lower bound (90% CI)	Analysis
ub90	Upper bound (90% CI)	Analysis
rn	Variable name (treatment indicator)	

Table 15: trustworthiness.csv — Exported marginal effects from the conjoint regression for trustworthiness

Variable Name	Description	Source
beta	Coefficient estimate	Analysis
se	Standard error	Analysis
lb95	Lower bound (95% CI)	Analysis
ub95	Upper bound (95% CI)	Analysis
lb90	Lower bound (90% CI)	Analysis
ub90	Upper bound (90% CI)	Analysis
rn	Variable name (treatment indicator)	

Table 16: aggressiveness.csv — Exported marginal effects from the conjoint regression for aggressiveness

Variable Name	Description	Source
beta	Coefficient estimate	Analysis
se	Standard error	Analysis
lb95	Lower bound (95% CI)	Analysis
ub95	Upper bound (95% CI)	Analysis
lb90	Lower bound (90% CI)	Analysis
ub90	Upper bound (90% CI)	Analysis
rn	Variable name (treatment indicator)	

Table 17: conjoint_image_data.csv — Metadata for conjoint experiment images

Variable Name	Description	Source
id	Row index	
path	Image file path	
aggressiveness	Model predicted aggressiveness score	Analysis
competence	Model predicted competence score	Analysis
trustworthiness	Model predicted trustworthiness score	Analysis
attractiveness	Model predicted attractiveness score (spelling as stored in source data)	Analysis
selection	Selection category	
control_diff	Average difference in scores of the three control dimensions	Analysis
target_diff	Difference in score of the target dimension	Analysis

Table 18: processed_conjoint_answer_user.csv — Raw conjoint experiment response data before processing

Variable Name	Description	Source
id	Response ID	Survey
user_id	Unique user identifier	Survey
a_image	Image A file path	Survey
b_image	Image B file path	Survey
a_info	Candidate A profile information	Survey
b_info	Candidate B profile information	Survey
answer	Respondent's choice (A or B)	Survey
starttime	Survey start time	Survey
finishtime	Survey finish time	Survey
answer_image	Selected image path	Survey
s_r_dimension	Dimension for selection	Survey
image	Image category	Survey
type	Experiment type	Survey
dimension	Facial trait dimension	Survey
score	Score	Survey
number	Trial number	Survey
a_birth_year	Candidate A birth year	Survey
a_education	Candidate A education level	Survey
a_experience	Candidate A work experience	Survey
b_birth_year	Candidate B birth year	Survey
b_education	Candidate B education level	Survey
b_experience	Candidate B work experience	Survey
id_y	Respondent ID	Survey
gender	Respondent gender	Survey
age	Respondent age group	Survey
dropdown	Respondent education	Survey
qualifications	Respondent province	Survey
career	Respondent career type	Survey
level	Respondent position level	Survey
Assessment	Assessment criteria	Survey
promotion	Promotion criteria	Survey
status	Response status	Survey

Table 19: conjoint.dta — Conjoint survey experiment data with respondent choices and randomized candidate profiles

Variable Name	Description	Source
<i>Identifiers</i>		
id	Response ID	Survey
user_id	Unique user identifier	Survey

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Variable Name	Description	Source
id_y	Respondent ID (from merge)	Survey
<i>Candidate Attributes</i>		
a_image	Image A file path	Survey
b_image	Image B file path	Survey
a_info	Candidate A profile information (raw string)	Survey
b_info	Candidate B profile information (raw string)	Survey
a_birth_year	Candidate A birth year	Survey
a_education	Candidate A education level	Survey
a_experience	Candidate A work experience	Survey
b_birth_year	Candidate B birth year	Survey
b_education	Candidate B education level	Survey
b_experience	Candidate B work experience	Survey
<i>Response Variables</i>		
answer	Respondent's choice (A or B)	Survey
answer_image	Image file path of the chosen option	Survey
choice	Long-format choice indicator (a_full / b_full)	Derived
chosen	1 if this row corresponds to the chosen candidate	Derived
image_id	Image side indicator (A or B)	Derived
<i>Experiment Design</i>		
image	Image category label	Survey
type	Experiment type	Survey
dimension	Facial trait dimension varied in this task	Survey
score	Score of the manipulated facial dimension	Survey
selection	Selection category (derived from image)	Survey
s_r_dimension	Dimension used for selection/rating	Survey
number	Trial number	Survey
<i>Facial Trait Ratings (Model-Predicted)</i>		
aggressiveness	Predicted aggressiveness score	Analysis
competence	Predicted competence score	Analysis
trustworthiness	Predicted trustworthiness score	Analysis
attractiveness	Predicted attractiveness score (spelling as stored in source data)	Analysis
control_diff	Average difference in scores of the three control dimensions	Analysis

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Variable Name	Description	Source
target_diff	Difference in score of the target dimension	Analysis
<i>Derived Long-Format Variables</i>		
info	Concatenated profile + image string for the candidate in this row	Derived
high	1 if the candidate in this row is the high-rated face	Derived
birthyear	Candidate birth year parsed from info	Derived
edu	Candidate education parsed from info (bachelor / master / PhD)	Derived
work	Candidate work experience parsed from info	Derived
<i>Respondent Demographics</i>		
gender	Respondent gender	Survey
age	Respondent age group	Survey
dropdown	Respondent education	Survey
qualifications	Respondent province	Survey
career	Respondent career type	Survey
level	Respondent position level	Survey
Assessment	Respondent's stated assessment criteria	Survey
promotion	Respondent's stated promotion criteria	Survey
status	Response status	Survey
<i>Timing</i>		
starttime	Survey start time	Survey
finishtime	Survey finish time	Survey

References

- Kazemi, V. and Sullivan, J. (2014). One millisecond face alignment with an ensemble of regression trees. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1867–1874.
- King, D. E. (2009). Dlib-ml: A machine learning toolkit. *Journal of Machine Learning Research*, 10:1755–1758.